

Talk Analysis:

Boston Infinite Money Glitch: Hacking Transit Cards Without Ending Up In Handcuffs

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Talk: DEF CON 31, August 2023, Zach Bertocchi, Scott Campbell, Noah Gibson, and Matthew Harris

Video: https://www.youtube.com/watch?v=1JT_1TfK69Q

1. The ask

Stated directly at 2:33–2:56: the audience should leave able to reverse-engineer a transit card, know how to disclose a vulnerability to a government agency without getting sued, and, failing that, have a fun story. The real ask is narrower. A system nobody has cracked in fifteen years is crackable by four vocational-school students with a \$9 NFC reader, provided you are patient enough to fail through three broken readers first.

2. The hero

They are the heroes of their own story, despite two early gestures the other way. At 1:37–1:53 they deny expertise (“we don’t actually have any real qualifications... we call ourselves amateur miscreants”), and at 2:33 they frame the payoff around what the audience gets. The narrative from 12:43 to 35:34 is still a first-person account of their own trial and error: three failed NFC readers, a soldering attempt, a 24-hour brute-force run that returned nothing. At 21:14 a successful clone draws a genuine round of applause. The room is applauding what they did, not something the audience can now do.

3. The shape

The sparkline starts at *what is*: a 2004 smartcard running proprietary 48-bit encryption that nobody has cracked in fifteen years (6:26–11:59). The first turning point comes at 16:34, when default keys turn up in a public repository after two failed attacks; that is where *what-could-be* first looks reachable. The second turning point, and the more important one, is the XOR breakthrough at 26:27, which turns cloning into forging and lets them push a card’s balance to \$327.67 before hitting a 16-bit integer overflow. The S.T.A.R. moment is the live demo at 43:20–44:04. Onstage they tap a \$0.25 card, edit its value to \$251.50 on the reader, and re-tap it to show the change took, then repeat the trick to reclassify it as an employee card.

4. What landed

“This wouldn’t be Defcon without computer glitches” (4:29) gets a laugh (4:34) and buys six more minutes of slide trouble. The 21:14 applause matters more. It follows nine minutes of narrated failure (three dead readers, a null 24-hour run), so the crowd is rewarding a payoff it watched get earned. Duarte’s *what-is/what-could-be* contrast only works if the audience believes *what-is* was hard, and the reader-buying saga supplies that.

5. What did not

At 36:07–36:15 a speaker says “we only got 10 minutes left,” and the explanation compresses from there. The vocabulary for the automated forging process (data modifier, checksum modifier, lookup table) is introduced and then dropped mid-sentence: “I’m just going to skip part of this” (36:46). The lookup table itself turns out to be something they were asked not to share (37:02–37:18), which cuts the technical payoff right when it should land. A second AV failure at 41:12–42:17 (“wrong one, wrong one... no, the other video”) means the S.T.A.R. moment has to recover from stage trouble twice.

6. One change

Cut the disclosure section (39:36–40:53, Bobby Rauch’s article and the MBTA complaint form) to two sentences and give the reclaimed minute to the XOR explanation. Disclosure is a what-should-happen-next argument and survives being told fast. XOR is the actual technical turn of the talk and the one section that runs out of time.